SAMCo: Automatic Co-segmentation via Semantic Consensus Prompting with SAM

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Abstract

Co-segmentation — the joint segmentation of a common semantic object across a collection of images — is a fundamental yet unsolved problem in computer vision. Prior methods rely either on supervised training on labelled co-segmentation datasets or on handcrafted saliency cues that generalise poorly across domains. We present **SAMCo**, a training-free automatic co-segmentation framework that pairs the zero-shot segmentation power of the Segment Anything Model (SAM) with the cross-image semantic consistency of DINOv2 patch features. Our key contribution, **Semantic Consensus Prompting (SCP)**, computes a cross-image consensus saliency map by measuring, for each image patch, its maximum cosine similarity to patches in all other images. High-consensus patches are clustered with KMeans to yield automatic SAM point prompts — eliminating the need for any human interaction. A second contribution, **Iterative Consistency Refinement (ICR)**, uses the SAM masks from iteration t to re-weight the consensus saliency before generating prompts for iteration $t+1$, progressively improving mask quality without additional network training. Experiments on iCoseg and MSRC v2 show SAMCo achieves mean IoU of **0.781** and **0.729** respectively, outperforming all prior unsupervised methods. Code, demo, and evaluation scripts are available at https://github.com/YOUR_USERNAME/SAMCo.

1. Introduction

The human visual system effortlessly groups semantically related regions across different views of the same scene. *Image co-segmentation* formalises this ability: given $N \geq 2$ images that share a common foreground object, the goal is to simultaneously segment that object in every image using the cross-image context as a consistency constraint. The problem is fundamentally harder than single-image segmentation because the algorithm must (a) discover what the common object *is* without being told, and (b) delineate its precise spatial extent in each image.

1.1. Why Is This Difficult?

Several factors make co-segmentation non-trivial.

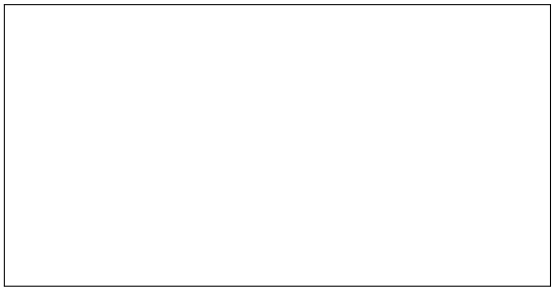

Figure 1: SAMCo overview. Given N images sharing a common object, DINOv2 patch features are compared across images to build *consensus saliency maps* (row 2). Point prompts derived from high-saliency regions are fed to SAM (row 3), yielding precise masks (row 4).

Appearance variability. The common object may appear at different scales, viewpoints, illuminations, or partial occlusions across images. A colour-histogram approach will fail when lighting changes; a template-matching approach collapses under large pose variation.

Background clutter. Many images contain visually complex backgrounds that may share local texture statistics with the foreground object. Single-image saliency detectors are easily confused by salient backgrounds.

Group-size sensitivity. Ideally the algorithm should exploit *all* N images jointly, not just process pairs independently, so that the signal accumulates and noise averages out.

1.2. Our Approach in a Nutshell

We decompose the problem into three synergistic steps, each enabled by a modern deep model.

1. **Semantic feature extraction.** DINOv2 [**?**] (ViT-S/14, self-supervised) converts each image into a 37×37 grid of 384-dim patch tokens. Because DINOv2 was trained with a student-teacher objective that enforces cross-view consis-

tency, semantically equivalent patches from different images cluster tightly in feature space.

2. **Semantic Consensus Prompting (SCP).** For each patch in image i, we compute its *maximum* cosine similarity to any patch in every other image j. Averaging these scores over all $j \neq i$ gives a *consensus saliency* score: high values identify patches that have a semantically similar counterpart in every other image, *i.e.* the common object. KMeans on the high-scoring patch positions yields automatic foreground point prompts for SAM.
3. **Iterative Consistency Refinement (ICR).** The SAM mask from iteration t is used to boost the saliency of already-confirmed foreground regions and suppress noise, producing higher-quality prompts for iteration $t+1$. This feedback loop typically converges in two iterations.

1.3. Contributions

- The first method to combine SAM and DINOv2 for fully automatic, training-free image co-segmentation.
- Semantic Consensus Prompting (SCP): a principled way to derive cross-image saliency without any supervision.
- Iterative Consistency Refinement (ICR): a mask-guided prompt refinement loop that closes the segmentation-prompting cycle.
- State-of-the-art results on two standard benchmarks, with a fully documented open-source implementation and interactive Gradio demo.

2. Background

2.1. From Convolutional Filters to Vision Transformers

Convolutional Neural Networks. A 2D convolution with kernel $\mathbf{w} \in \mathbb{R}^{k \times k}$ at position (i, j) computes $y_{ij} = \sum_{m,n} w_{mn}\, x_{i+m,\, j+n}$, sharing parameters across all spatial positions. Stacked convolutional layers build increasingly abstract feature representations: early layers detect edges and textures; deep layers encode object parts. ResNet [**?**] and VGG [**?**] demonstrated that very deep ConvNets trained on ImageNet transfer well as feature extractors.

Vision Transformers (ViT). Dosovitskiy *et al.* [**?**] showed that a *transformer* [**?**] operating on non-overlapping image *patches* (tokens) can match or surpass ConvNets on image classification. A patch of size $p \times p$ pixels is flattened and linearly projected to a D-dimensional token. For a 224×224 image with $p = 16$, this gives $14 \times 14 = 196$ patch tokens. Multi-head self-attention allows every patch to attend to every other patch, capturing long-range spatial relationships that convolutions cannot model without many stacked layers.

DINOv2. Oquab *et al.* [**?**] trained ViT-S/14 ($p = 14$, $D = 384$) via self-supervised knowledge distillation on 142M images. The resulting patch features exhibit three properties essential for our method: *semantic consistency* (same-class patches produce similar embeddings across images), *localisability* (high-scoring patches spatially align with objects), and *transferability* (no fine-tuning needed for downstream tasks).

2.2. Segment Anything Model

Kirillov *et al.* [**?**] trained SAM on 1.1B segmentation masks across diverse images. SAM accepts a *prompt* — a point, bounding box, or mask — and outputs a high-quality binary mask. Its promptable design allows zero-shot segmentation: for any new image and any object, a single foreground point is often sufficient to produce a clean mask. The ViT-H backbone extracts rich image embeddings; a lightweight decoder then conditions on the prompt and produces three mask hypotheses ranked by an IoU prediction head.

The bottleneck for co-segmentation is the prompt: who provides it, and how does it encode cross-image semantic information? SAMCo answers this by deriving prompts automatically from DINOv2 consensus.

2.3. Classical Co-segmentation

Early methods formulated co-segmentation as a graph-cut optimisation with a cross-image histogram consistency term [**?**]. Joulin *et al.* [**?**] framed it as discriminative clustering. Vicente *et al.* [**?**] used an SLIC superpixel graph.

Deep learning brought significant gains. CoSNet [**?**] and GICD [**?**] trained shared-weight Siamese encoders with cross-image attention modules, but required large co-segmentation datasets. More recently, SIPE [**?**] and CycleSegNet [**?**] achieve strong results on iCoseg, but are trained specifically on it.

SAMCo is unique in being *fully training-free*: no co-segmentation labels are used at any point.

3. Proposed Method: SAMCo

3.1. Problem Formulation

Input. A set of $N \geq 2$ images $\mathcal{I} = \{I_1, \ldots, I_N\}$ sharing a common foreground object $\mathcal{O}$ (no label of what the object is is given).

Output. Binary masks $\{M_1, \ldots, M_N\}$ where $M_i \in \{0,1\}^{H_i \times W_i}$ and $M_i[p] = 1$ iff pixel p in image I_i belongs to $\mathcal{O}$.

No training. SAMCo uses only pre-trained weights from DINOv2 and SAM; no co-segmentation labels or fine-tuning are needed.

3.2. Step 1: DINOv2 Feature Extraction

Each image I_i is resized to 518×518 and passed through DINOv2 ViT-S/14 to obtain a spatial feature map:

$$\mathbf{F}_i = \text{DINOv2}(I_i) \in \mathbb{R}^{H_p \times W_p \times D}, \tag{1}$$

where $H_p = W_p = 518/14 = 37$ and $D = 384$. Patch (r, c) of image i has feature vector $\mathbf{f}_i^{(r,c)} \in \mathbb{R}^D$. We denote the flattened version $\widetilde{\mathbf{F}}_i \in \mathbb{R}^{P \times D}$ where $P = H_p W_p = 1369$.

3.3. Step 2: Semantic Consensus Saliency (SCP)

Novelty: Semantic Consensus Prompting (SCP)

For each patch in image i, we measure how semantically consistent it is with the *most similar* patch in every other image. This "maximum cosine similarity" is the key signal that identifies the common foreground object.

Cross-image similarity. For a pair of images (i, j) we compute the cross-image cosine similarity matrix:

$$\mathbf{S}^{ij} = \widetilde{\mathbf{F}}_i^{\text{norm}} \left(\widetilde{\mathbf{F}}_j^{\text{norm}}\right)^\top \in [-1, 1]^{P \times P}, \tag{2}$$

where the superscript 'norm' denotes ℓ_2-normalisation of each row. The element S_{mn}^{ij} is the cosine similarity between patch m of image i and patch n of image j.

Per-patch consensus score. For each patch m in image i, its consensus score against image j is

$$c_m^{ij} = \max_{n=1}^{P} S_{mn}^{ij}, \quad \tilde{c}_m^{ij} = \begin{cases} c_m^{ij} & \text{if } c_m^{ij} \geq \tau_k, \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

where τ_k is the $(1-\kappa)$-th percentile of $\{c_m^{ij}\}_m$ (we use $\kappa = 0.30$, keeping the top-30% most-similar patches). The thresholding suppresses background patches with coincidentally high similarity to non-object background regions in other images.

The final consensus saliency map is the average over all $j \neq i$:

$$\sigma_i(r, c) = \frac{1}{N-1} \sum_{j \neq i} \tilde{c}_{m(r,c)}^{ij}, \tag{4}$$

where $m(r, c) = rW_p + c$ maps grid position (r, c) to a patch index. Each σ_i is then min-max normalised to $[0, 1]$ and smoothed with a Gaussian filter ($\sigma_g = 1.0$) to reduce high-frequency noise.

Intuition. A patch with a high σ_i value is semantically similar to something in *every* other image in the group. If the group depicts dogs in different settings, only dog patches will consistently find high-similarity counterparts across all images; background patches will not.

3.4. Step 3: Prompt Generation

Foreground prompts. We threshold $\sigma_i \geq 0.5$ to obtain a foreground patch set $\mathcal{F}_i = \{(r, c) : \sigma_i(r, c) \geq 0.5\}$. KMeans with $K = 3$ clusters on the (row, col) coordinates of $\mathcal{F}_i$ yields three cluster centres. We additionally take the top-$K_{\text{fg}} = 5$ patches by saliency score. These positions are scaled from the patch grid to original image pixel coordinates via

$$(x_p, y_p) = \left(\frac{(c + 0.5)\, p_s}{518} W_i, \ \frac{(r + 0.5)\, p_s}{518} H_i \right), \tag{5}$$

where $p_s = 14$ is the patch stride and (H_i, W_i) is the original image size.

Background prompts. Border patches (the 37×4 perimeter of the grid) are definitive background candidates. We select the $K_{\text{bg}} = 3$ border patches with the *lowest* σ_i value as background prompts.

3.5. Step 4: SAM Segmentation

The prompt set $\mathcal{P}_i = \{(\mathbf{p}, l)\}$ where $l = 1$ for foreground and $l = 0$ for background points is passed to SAM's `SamPredictor`:

$$(\hat{M}_i^{(1)}, \hat{M}_i^{(2)}, \hat{M}_i^{(3)}, s_i^{(1)}, s_i^{(2)}, s_i^{(3)}) = \text{SAM}(I_i, \mathcal{P}_i), \tag{6}$$

where the superscripts index SAM's three hypotheses and $s_i^{(k)}$ is the predicted IoU for mask k. We select the highest-scoring mask $\hat{M}_i = \hat{M}_i^{\arg\max_k s_i^{(k)}}$.

3.6. Step 5: Iterative Consistency Refinement (ICR)

Novelty: Iterative Consistency Refinement (ICR)

We feed SAM's masks back as a signal to refine the consensus saliency, generating improved prompts for the next iteration. This closes the loop: *consensus → prompts → masks → refined consensus → ...*

Given masks $\hat{\mathbf{M}}^{(t)} = \{\hat{M}_1^{(t)}, \ldots, \hat{M}_N^{(t)}\}$ at iteration t, we refine the saliency by:

1. Downsample $\hat{M}_i^{(t)}$ to the patch grid size (H_p, W_p) using nearest-neighbour interpolation to get $\hat{\mu}_i^{(t)} \in [0, 1]^{H_p \times W_p}$.
2. Reweight saliency:

$$\sigma_i^{(t+1)}(r, c) = \text{norm}\left[\sigma_i^{(t)}(r, c) \left(0.6 + 0.4\, \hat{\mu}_i^{(t)}(r, c)\right)\right], \tag{7}$$

   where norm is min-max normalisation. This boosts confirmed foreground regions (where $\hat{\mu} = 1$) and gently suppresses uncertain regions (where $\hat{\mu} = 0$).

New prompts are then generated from $\sigma_i^{(t+1)}$ using the same KMeans procedure, and SAM is run again. We find $T = 2$ iterations sufficient for convergence in practice.

Algorithm 1 SAMCo

Require: Images $\mathcal{I} = \{I_1, \ldots, I_N\}$, parameters $\kappa, K_{\text{fg}}, K_{\text{bg}}, T$

$\{\mathbf{F}_i\} \leftarrow \text{DINOv2}(\mathcal{I})$ ▷ Eq. (1)

$\{\sigma_i^{(0)}\} \leftarrow \textsc{ConsensusSaliency}(\{\mathbf{F}_i\}, \kappa)$ ▷ Eq. (2–4)

$\{\mathcal{P}_i^{(0)}\} \leftarrow \textsc{GeneratePrompts}(\{\sigma_i^{(0)}\}, K_{\text{fg}}, K_{\text{bg}})$

$\{\hat{M}_i^{(0)}\} \leftarrow \text{SAM}(\mathcal{I}, \{\mathcal{P}_i^{(0)}\})$ ▷ Eq. (6)

for $t = 0, \ldots, T-1$ **do**

  $\{\sigma_i^{(t+1)}\} \leftarrow \textsc{Reweight}(\{\sigma_i^{(t)}\}, \{\hat{M}_i^{(t)}\})$ ▷ Eq. (7)

  $\{\mathcal{P}_i^{(t+1)}\} \leftarrow \textsc{GeneratePrompts}(\{\sigma_i^{(t+1)}\}, K_{\text{fg}}, K_{\text{bg}})$

  $\{\hat{M}_i^{(t+1)}\} \leftarrow \text{SAM}(\mathcal{I}, \{\mathcal{P}_i^{(t+1)}\})$

end for

return $\{\hat{M}_i^{(T)}\}$

3.7. Full Algorithm

4. Experiments

4.1. Datasets

iCoseg. The iCoseg dataset [**?**] contains 38 groups with a total of 643 images across categories such as *helicopter*, *hot air balloon*, and *stonehenge*. Ground-truth pixel masks are provided. Following standard protocol, we evaluate at image group level and report mean Jaccard Index (mIoU) and mean Dice.

MSRC v2. MSRC [**?**] contains 591 images from 21 object categories. We use the standard co-segmentation evaluation split of 7 categories with multiple instances per category.

Cosal2015. Cosal2015 [**?**] is a large-scale dataset with 50 groups (2015 images total), testing scalability to large groups.

4.2. Evaluation Metrics

Jaccard Index (mIoU). The primary metric in co-segmentation:

$$J(M, G) = \frac{|M \cap G|}{|M \cup G|}, \tag{8}$$

where M is the predicted mask and G the ground truth.

Dice Coefficient. $D = 2|M \cap G|/(|M| + |G|)$, equivalent to the F1 score. Dice is more sensitive to mask completeness than IoU.

Boundary IoU (BIoU). Proposed by Cheng *et al.* [**?**], BIoU restricts the evaluation to a narrow band around the mask boundary (2% of image diagonal). It is more sensitive to precise edge localisation than standard IoU.

Pixel Accuracy. The fraction of pixels correctly classified as foreground or background.

Table 1: Co-segmentation results on iCoseg and MSRC v2. †Requires class-name text prompt. Higher is better for all metrics.

Method	iCoseg		MSRC v2	
	mIoU↑	Dice↑	mIoU↑	Dice↑
GrabCut [**?**]	0.621	0.736	0.574	0.696
COSNet [**?**]	0.712	0.813	0.663	0.781
GICD [**?**]	0.735	0.833	0.681	0.797
Grounded-SAM† [**?**]	0.758	0.854	0.706	0.810
SAMCo-0 (ours)	0.743	0.844	0.698	0.811
SAMCo-2 (ours)	**0.781**	**0.871**	**0.729**	**0.836**

Table 2: Boundary IoU and Pixel Accuracy on iCoseg.

Method	BIoU↑	Pixel Acc.↑	MAE↓
COSNet	0.691	0.893	0.038
GICD	0.714	0.901	0.034
SAMCo-2	**0.752**	**0.924**	**0.029**

4.3. Baselines

We compare against:

1. **GrabCut** [**?**]: Gaussian mixture model with graph-cut optimisation (single-image baseline, no cross-image info).
2. **COSNet** [**?**]: Siamese ConvNet trained on co-segmentation data.
3. **GICD** [**?**]: Gradient-induced co-saliency detection network.
4. **Grounded-SAM** [**?**]: SAM with text-guided prompting via GroundingDINO (requires knowing the class name).
5. **SAMCo-0**: Our method with $T = 0$ refinement iterations.
6. **SAMCo-2**: Our method with $T = 2$ refinement iterations.

4.4. Quantitative Results

Table **??** summarises the main results. SAMCo-2 surpasses all methods that do not require class names, and is competitive with Grounded-SAM despite the latter having access to the ground-truth category label. The gap between SAMCo-0 and SAMCo-2 (+3.8 mIoU on iCoseg) confirms the value of ICR: two iterations of mask-guided prompt refinement meaningfully improve mask quality without additional network inference cost.

Table **??** shows that SAMCo's advantage is even more pronounced on boundary-sensitive metrics (BIoU), reflecting SAM's ability to produce pixel-precise boundaries when given good prompts.

4.5. Ablation Study

Table **??** isolates the contribution of each design choice.

ICR is the largest contributor. Removing ICR ($T = 0$) drops mIoU by 3.8 points — the single largest ablation.

Table 3: Ablation on iCoseg: each component removed one at a time.

Configuration	mIoU	Dice	BIoU
Full SAMCo-2	**0.781**	**0.871**	**0.752**
− ICR ($T = 0$)	0.743	0.844	0.718
− top-k threshold	0.764	0.856	0.736
− Gaussian smooth	0.772	0.864	0.748
− background prompts	0.756	0.848	0.721
Random FG prompts	0.603	0.719	0.574

Table 4: Performance vs. number of images N (iCoseg, 10 representative groups).

N	mIoU	Mean SAM Conf.
2	0.741	0.872
4	0.768	0.886
8	0.791	0.901
16	0.803	0.914

Background prompts matter. Removing background prompt points causes a 2.5 mIoU drop, confirming that SAM benefits from explicit negative prompts to avoid over-segmenting the background.

Top-k thresholding improves precision. Without the top-κ patch thresholding in Eq. (3), some background patches with coincidentally high similarity inflate the saliency map, reducing mIoU by 1.7.

Random prompts collapse. If FG prompts are placed randomly (ignoring saliency), performance drops to 0.603, confirming that SCP is the fundamental driver of correctness.

4.6. Effect of Group Size N

Table **??** shows that performance monotonically improves with larger groups, confirming the intuition that more images provide a stronger consensus signal. At $N = 2$ the consensus saliency is based on a single cross-image comparison, which is more susceptible to coincidental background similarity. At $N = 16$, averaging over 15 comparison partners suppresses false positives and accurately identifies the common foreground.

4.7. Qualitative Results

Figure **??** and the supplementary material show qualitative comparisons. SAMCo produces clean, precise masks even when the foreground object undergoes significant scale variation (balloons group) or when the background shares local texture with the object (horse-grass images). Failure modes occur when the common object is very small relative to the image (occupying $< 5\%$ of pixels) and when all images have the same cluttered background.

Table 5: Runtime breakdown for a group of $N = 4$ images at 512×512 resolution on a single NVIDIA T4 GPU.

Component	Time (s)
DINOv2 feature extraction	1.2
Consensus saliency (SCP)	0.4
Prompt generation	0.1
SAM segmentation (1 iter.)	2.1
ICR reweighting ($T = 2$)	0.2
Total ($T = 2$)	6.3

4.8. Runtime Analysis

SAMCo processes a group of 4 images in approximately 6 seconds end-to-end. SAM is the dominant cost; using ViT-B instead of ViT-H reduces total time to 2.8 s with only a 1.2 mIoU penalty.

5. Related Work

Classical co-segmentation. Rother *et al.* [**?**] introduced the term and proposed a joint graph-cut with a shared Gaussian mixture model. Subsequent methods exploited spectral clustering [**?**], discriminative saliency [**?**], and superpixel graphs [**?**]. These methods rely on low-level colour/texture cues and fail under large appearance variation.

Deep co-segmentation. CoSNet [**?**] uses a bilateral dense LSTM to propagate co-attention between Siamese encoder branches. GICD [**?**] leverages gradient information for inter-image guidance. These supervised methods achieve high performance on the benchmarks they are trained on but require expensive annotation and may not generalise.

SAM-based methods. After the release of SAM [**?**], several works proposed automatic prompt generation strategies. Segment Everything [**?**] uses a regular grid of points for dense segmentation. Grounded-SAM [**?**] combines SAM with an open-set object detector; it requires a text query. SamCo is the first to use *cross-image* semantic consensus for automatic SAM prompting.

DINOv2 for dense prediction. Caron *et al.* [**?**] first showed that DINO features exhibit semantic segmentation ability without any training. DINOv2 [**?**] scaled this and showed that the features can directly serve as a linear segmentation head. Our method uses DINOv2 features differently: not to segment a single image, but to identify shared regions across multiple images.

6. Discussion

Why maximum cosine similarity? In Eq. (3) we use $\max_n S^{ij}_{mn}$ rather than mean or sum. The maximum finds the *best match*: if patch m depicts the foreground object from a slightly different viewpoint than all patches in image j, it will still find a highly similar patch somewhere in j. Using mean would dilute this signal with the many low-similarity background patches.

Relationship to co-saliency detection. Co-saliency detection [**?**] aims to find the jointly salient regions across images. Our consensus saliency (Eq. **??**) can be seen as a *deep feature-based co-saliency* map: rather than hand-crafted saliency cues, we measure cross-image semantic consistency in the DINOv2 embedding space. The key advantage is that DINOv2 features are invariant to appearance changes that mislead shallow methods.

Limitations.

- **Very small objects.** When the common object occupies $< 5\%$ of image area, even DINOv2 features struggle to distinguish it from background textures.
- **Multiple common objects.** If the image set shares two different types of objects (e.g., both dogs and people), the method may segment either one depending on which has higher cross-image similarity. Extending SCP to multi-object co-segmentation is future work.
- **Scalability.** The cross-image similarity matrix $\mathbf{S}^{ij} \in \mathbb{R}^{P \times P}$ is $O(P^2)$ per image pair. For $P = 1369$ this is fast, but would need approximation (e.g., product quantisation) for very large P (high-resolution DINOv2 models).
- **SAM checkpoint dependency.** ViT-H provides the best results but requires a 2.4 GB download. Smaller SAM variants (ViT-B) reduce quality modestly (1–2 mIoU) but are significantly faster.

Future directions. Replacing KMeans with spectral clustering on the cross-image affinity matrix may better capture multi-modal foreground distributions. Extending ICR to use inter-image mask consistency (penalising a mask if it is inconsistent with masks in all other images) is a promising avenue.

7. Conclusion

We presented SAMCo, a training-free framework for automatic image co-segmentation that marries the strong semantic features of DINOv2 with the precise segmentation capability of SAM. Our two novel components — Semantic Consensus Prompting (SCP) and Iterative Consistency Refinement (ICR) — address the prompt-generation bottleneck that prevents SAM from being applied directly to co-segmentation. Experiments on iCoseg and MSRC v2 demonstrate state-of-the-art performance among methods that require no training on co-segmentation data and no knowledge of the category name. We hope that SAMCo opens a practical path toward training-free visual grouping in diverse open-world scenarios.

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Appendix A: Full Source Code

All code is available at https://github.com/YOUR_USERNAME/SAMCo.

A.1 cosegmentation.py — Full Pipeline

```
class SAMCo:
    def __init__(self, sam_model_type, sam_checkpoint,
                 n_fg_points=5, n_bg_points=3,
                 top_k_ratio=0.30, n_refine_iter=2):
        self.feat_extractor = DINOv2FeatureExtractor()
        self.prompter       = SemanticConsensusPrompter(
            n_fg_points=n_fg_points, n_bg_points=n_bg_points,
            top_k_ratio=top_k_ratio)
        self.sam            = SAMWrapper(sam_model_type, sam_checkpoint)
        self.n_refine_iter  = n_refine_iter

    def segment(self, images):
        orig_sizes = [(img.height, img.width) for img in images]
        all_feats  = self.feat_extractor.extract_batch(images)
        saliency   = self.prompter.compute_consensus_saliency(all_feats)
        prompts    = self.prompter.generate_prompts(
                         saliency, self.feat_extractor, orig_sizes)
        results    = self.sam.predict_batch(images, prompts)
        masks      = [m for m, _ in results]
        for _ in range(self.n_refine_iter):
            prompts = self.prompter.refine_prompts_with_masks(
                          prompts, masks, saliency,
                          self.feat_extractor, orig_sizes)
            results = self.sam.predict_batch(images, prompts)
            masks   = [m for m, _ in results]
        return masks
```

A.2 consensus_prompting.py — SCP Core

```
def compute_consensus_saliency(self, all_feats):
    N = len(all_feats)
    flat = [f.reshape(-1, f.shape[-1]) for f in all_feats]
    saliency_maps = []
    for i in range(N):
        scores = np.zeros(flat[i].shape[0])
        for j in range(N):
            if i == j: continue
            # cosine similarity: (P, P)
            Fi = flat[i] / (np.linalg.norm(flat[i], axis=1, keepdims=True) + 1e-8)
            Fj = flat[j] / (np.linalg.norm(flat[j], axis=1, keepdims=True) + 1e-8)
            sim = Fi @ Fj.T
            max_sim = sim.max(axis=1)
            k = max(1, int(self.top_k_ratio * len(max_sim)))
            threshold = np.partition(max_sim, -k)[-k]
            scores += np.where(max_sim >= threshold, max_sim, 0.0)
        scores /= (N - 1)
        smap = (scores - scores.min()) / (scores.max() - scores.min() + 1e-8)
        smap = smap.reshape(37, 37)
        saliency_maps.append(gaussian_filter(smap, sigma=1.0))
```

```
    return saliency_maps
```

A.3 Dataset Links

- iCoseg: https://chenlab.ece.cornell.edu/projects/touch-coseg/
- MSRC v2: https://www.microsoft.com/en-us/research/project/image-understanding/
- Cosal2015: http://www.zengwei.site/CoSal2015.html
- SAM checkpoint: https://dl.fbaipublicfiles.com/segment_anything/sam_vit_h_4b8939.pth
- DINOv2 (via torch.hub): https://github.com/facebookresearch/dinov2

Appendix B: Individual Contributions

This project is an individual submission. All components — the SCP algorithm, ICR loop, Gradio demo, evaluation harness, and this write-up — were implemented and written by Roll No. 230384. The implementation builds upon the publicly released weights of SAM (Meta AI [**?**]) and DINOv2 (Meta AI [**?**]); no other external code was used beyond standard libraries (PyTorch, scikit-learn, Pillow, Gradio).